Supplementary Materials for Communication-Aware Time-Scale-Separated Bi-Level Coordination for USV-AUV Collaboration in Underwater Mobile Computing

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***Abstract*—This supplement reports additional simulations, numerical diagnostics, controlled comparisons, and outdoor prototype experiments for the communication-aware time-scale-separated USV–AUV coordination framework developed in the accompanying manuscript. Beyond the principal 2–4-AUV TD3/DSAC-T comparisons in the main paper, the analyses cover a formulation-level comparison with earlier FIM–RL and related coordination paradigms; long-horizon error evolution over 10,000 steps; determinant-equivalent FIM spectral conditioning and perturbation sensitivity for 2–10 AUVs; candidate-level agreement between the leader surrogate and system return; five component-wise ablations; comparisons under matched USV update intervals and travel budgets; robustness to correlated and bursty reception, outages, and acoustic-parameter changes; and three repeated outdoor trials per coordination method. We distinguish simulation-only diagnostics, controlled geometry tests, and field-observed navigation behavior, while maintaining consistent metric definitions and clearly distinguishing the evaluation protocols underlying the reported results.**

I. Scope and Evaluation Protocols

The accompanying manuscript formulates a USV-mediated underwater mobile computing system in which a surface leader optimizes localization-support geometry and motion on a slower time scale, while reinforcement-learning AUV followers act at every step using communication-aware information. Its main experimental section evaluates the integrated method against a synchronous FIM-guided reference with 2–4 AUVs and TD3/DSAC-T followers. This supplement expands the evidence along three complementary directions: prolonged and more challenging simulated operation, mechanism-isolating controlled comparisons, and physical navigation trials. It also provides a formulation-level comparison of the proposed coordination problem with the earlier FIM–RL framework and related asynchronous or communication-aware paradigms. All references to the main manuscript below refer to the accompanying revised paper, and supplementary figure, table, and equation numbering begins with the prefix S.

The additional simulation analyses reuse the system concepts and metric conventions of Section VI of the main manuscript. Unless a subsection states otherwise, the nominal workspace is $200\,\mathrm{m} \times 200\,\mathrm{m}$, the depth reference is 100 m, the carrier frequency is 20 kHz, the control-plane packet length is 4096 bits, and the leader update interval is $N_u = 5$ follower steps. Sampling and processing delays are respectively $\mathcal{U}(0, 0.333)$ s and 0.1 s, with an adopted acoustic sound speed of 1500 m/s. These settings describe a coordination-oriented control-plane simulator, not a fully resolved acoustic modem, time-stamped packet-arrival process, or medium-access protocol. Prior analyses of underwater acoustic propagation and network-induced constraints motivate distinguishing this abstraction from a fully specified communication network [1], [2]. In the nominal model, geometry-dependent communication terms use planar USV–AUV separation as an approximation; physical acoustic exchanges depend on three-dimensional slant range and additional hardware delays. Table S1 summarizes the additional evaluations and makes their distinct sampling or repeat structures explicit.

II. Long-Horizon Tracking and State-Estimation Robustness

A. Error Metrics and Coordination Mechanisms

The nominal 1,000-step simulation comparisons in the main manuscript establish a consistent tracking and USV-motion advantage under the evaluated delay-and-loss model. To examine behavior over longer periods, we consider a 10,000-step comparison between the Proposed and Stale-only configurations. This study separates three quantities that should not be conflated. For AUV k, the buffered-state mismatch is

$$e_{k,t}^{\mathrm{buf}} = \|\hat{\mathbf{p}}_{k,t} - \mathbf{p}_{k,t}\|_2, \tag{S1}$$

where $\hat{\mathbf{p}}_{k,t}$ is the position retained at the leader and $\mathbf{p}_{k,t}$ is the corresponding simulated actual position. The one-step response discrepancy is

$$e_t^{\mathrm{pred}} = \left\| \widetilde{\mathbf{P}}_{a,t+1}^{\mathrm{br}} - \mathbf{P}_{a,t+1}^{\mathrm{br}} \right\|_2, \tag{S2}$$

where the tilde denotes the forecast made for a leader update and the second term denotes the realized follower configuration. The third metric is the closed-loop tracking error used in Section VI of the main manuscript. The buffered and prediction errors describe distinct internal information mismatches rather than alternative measurements of tracking accuracy. Prediction uncertainty is also a recognized concern in model-based RL [3]; the present diagnostic measures

TABLE S1
ADDITIONAL EVALUATION PROTOCOLS AND THEIR RELATIONSHIP TO THE MAIN MANUSCRIPT.

Analysis	Evaluation protocol	Interpretation
Long-horizon error evolution	10,000 steps; 3,000/5,000/10,000-step checkpoints summarize the preceding 200 steps; ten simulation profiles.	Finite-horizon error diagnostics.
FIM geometric scaling	2, 4, 6, 8, and 10 AUVs; 1,000 configurations per size; three coordinate perturbations per configuration.	Fixed-geometry numerical diagnostics, not a ten-AUV trained deployment.
Candidate-ranking agreement	2–4 AUV teams; five-step cycle returns; separate 36-decision-snapshot action audit.	Agreement of candidate rankings, not an FIM-only causal experiment.
Component ablation	One USV, two AUVs, TD3; 1,000 steps; 20 independent repeats each for Full and five ablated variants under the same statistical protocol.	Conditional contributions of five mechanisms.
Matched-mobility comparison	Two-AUV TD3; 1,000 steps; common travel caps of 25–100 m, matched update intervals and a 250-step terminal window.	Tracking under controlled motion allowance.
Communication stress tests	2–4 AUVs; correlated/GE reception, paired trace replay, forced outages, acoustic sweeps and shared-service caps.	Controlled simulation evidence of coordination robustness under specified communication impairments.
Outdoor navigation	One custom USV, three custom AUVs; 60-s executions; three trials per method.	Physical, indirect navigation and motion evidence.

one-step follower-response discrepancy without adopting that work's dropout or trajectory-sampling algorithm.

Two features of the coordination procedure explain the motivation for these diagnostics. Successfully received control-plane observations update the USV's buffered information, while packet loss retains the previous buffer. In addition, the one-step follower response is reconstructed from the latest leader-accessible configuration at every scheduled leader update; the prior cycle's forecast is not recursively propagated as the initial condition for the next cycle. The conditional expected-error bound in Eq. (65) and stage-wise prediction-sensitivity result in Theorem V.5 of the main manuscript characterize these mechanisms under their respective assumptions. Neither mechanism implies perfect recovery through an arbitrary loss burst or unconditional closed-loop stability.

B. Checkpoint Results over 10,000 Steps

For the reported 3,000-, 5,000-, and 10,000-step checkpoints, each statistic summarizes the preceding 200-step window (respectively steps 2,800–2,999, 4,800–4,999, and 9,800–9,999 under zero-based indexing). Table S2 presents the mean $\pm$ standard deviation across the ten simulation profiles used for the checkpoint comparison.

For the Proposed configuration, the tracking-error means are 0.251, 0.247, and 0.244 m, compared with 0.299, 0.291, and 0.291 m for Stale-only. The final-window reduction is approximately 16.2%. Buffered-state means remain between 1.630 and 1.651 m for Proposed, whereas Stale-only records 2.151, 2.116, and 2.223 m. At the final checkpoint the relative difference is approximately 26.3%. The corresponding one-step response errors are 1.026, 1.021, and 1.097 m for Proposed and 1.455, 1.541, and 1.639 m for Stale-only; the final reduction is approximately 33.1%.

Figs. S1 and S2 give complementary temporal and checkpoint views. The Proposed profile does not exhibit a sustained increase in the three checkpoint tracking-error means, although its one-step prediction error rises modestly in the final window. The results are therefore consistent with the intended buffered-refresh and nonrecursive one-step reconstruction mechanisms in the tested horizon. They do not rule out rare excursions between checkpoint windows, establish formal learned-policy stability, or identify the independent causal contribution of each mechanism. For Stale-only, which does not execute an online response-prediction module, the one-step discrepancy is used only as a diagnostic comparison at matched indices; it is not interpreted as an error output by an online predictor in that baseline.

III. FIM Scalability and Numerical Sensitivity

A. Determinant-Equivalent Fixed-Order Representation

The FIM study extends the four-AUV maximum team size of the main closed-loop evaluations by examining the original leader-side scalar information criterion under controlled geometries. This geometry-level focus is consistent with the broader literature on the optimality of sensor–target localization configurations [4], without treating the implemented determinant as a separately validated physical measurement model. The USV still optimizes a two-dimensional position. Although the AUV team has $2M$ planar position coordinates, the scalar determinant implemented in the response admits an equivalent 2×2 symmetric representation,

$$S_i = \sqrt{\|\mathbf{p}_i - \mathbf{p}_u\|_2^2 + h_u^2}, \qquad b_M = \sum_{i=1}^{M} S_i^{-2},$$
$$A_i = \frac{\|\mathbf{p}_i\|_2^4 - 2S_i^2\|\mathbf{p}_i\|_2^2}{2S_i^6}, \qquad \mathbf{u}_i = \frac{\mathbf{p}_i - \mathbf{p}_u}{\|\mathbf{p}_i - \mathbf{p}_u\|_2}, \tag{S3}$$

$$\mathbf{J}_M^{\mathrm{eff}} = b_M \mathbf{I}_2 + 2\sum_{i=1}^{M} A_i \mathbf{u}_i \mathbf{u}_i^{\mathsf{T}}. \tag{S4}$$

For the angle θ_{ij} between $\mathbf{u}_i$ and $\mathbf{u}_j$, its determinant equals the original scalar criterion:

$$\det(\mathbf{J}_M^{\mathrm{eff}}) = b_M\left(b_M + 2\sum_i A_i\right) + 4\sum_{i<j} A_i A_j \sin^2\theta_{ij}. \tag{S5}$$

The equivalent matrix is used only as a numerical diagnostic of the implemented determinant. In particular, its eigenvalues

TABLE S2
LONG-HORIZON CHECKPOINT COMPARISON (MEAN ± STANDARD DEVIATION (SD) ACROSS TEN SIMULATION PROFILES; PRECEDING 200-STEP WINDOWS).

Metric	Configuration	3,000 steps	5,000 steps	10,000 steps
Tracking error (m)	Proposed	$\mathbf{0.251 \pm 0.026}$	$\mathbf{0.247 \pm 0.039}$	$\mathbf{0.244 \pm 0.024}$
	Stale-only	0.299 ± 0.036	0.291 ± 0.038	0.291 ± 0.037
Buffered-state error (m)	Proposed	$\mathbf{1.651 \pm 0.167}$	$\mathbf{1.630 \pm 0.212}$	$\mathbf{1.639 \pm 0.334}$
	Stale-only	2.151 ± 0.446	2.116 ± 0.493	2.223 ± 0.565
One-step response error (m)	Proposed	$\mathbf{1.026 \pm 0.149}$	$\mathbf{1.021 \pm 0.182}$	$\mathbf{1.097 \pm 0.226}$
	Stale-only	1.455 ± 0.321	1.541 ± 0.237	1.639 ± 0.308

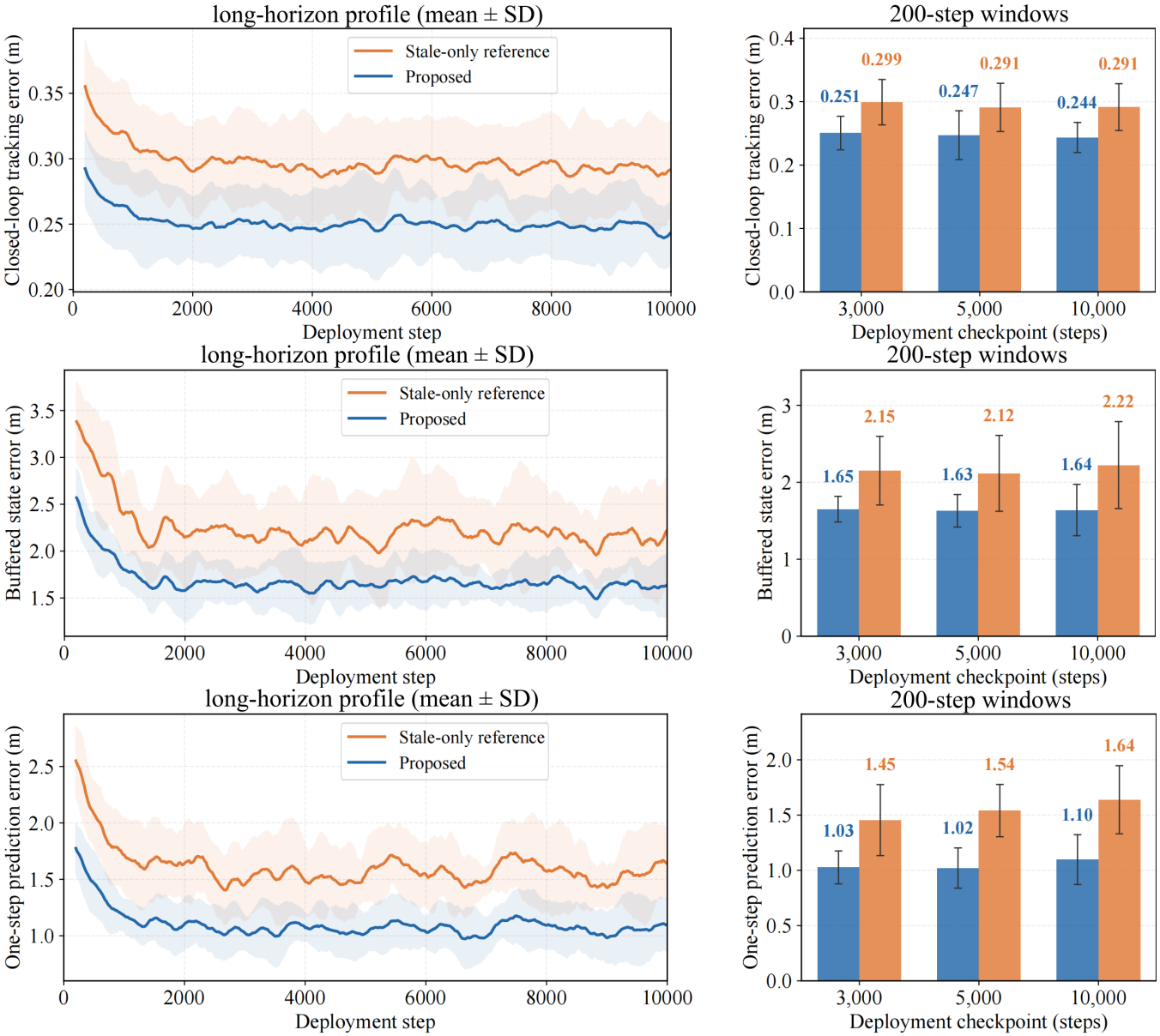


Fig. S1. Long-horizon tracking, buffered-state, and one-step follower-response error profiles, together with the corresponding 200-step checkpoint summaries. The left panels show 10,000-step simulation profiles; the right panels summarize means and standard deviations over the preceding 200-step windows.

are not presented as the spectrum of a different, growing-dimensional $2M \times 2M$ joint-state FIM. The equality does not independently validate a physical measurement FIM or establish invariance of this implemented scalar score to changes in the global coordinate origin; the latter warrants separate scrutiny because A_i contains $\|\mathbf{p}_i\|_2$ in the adopted coordi-

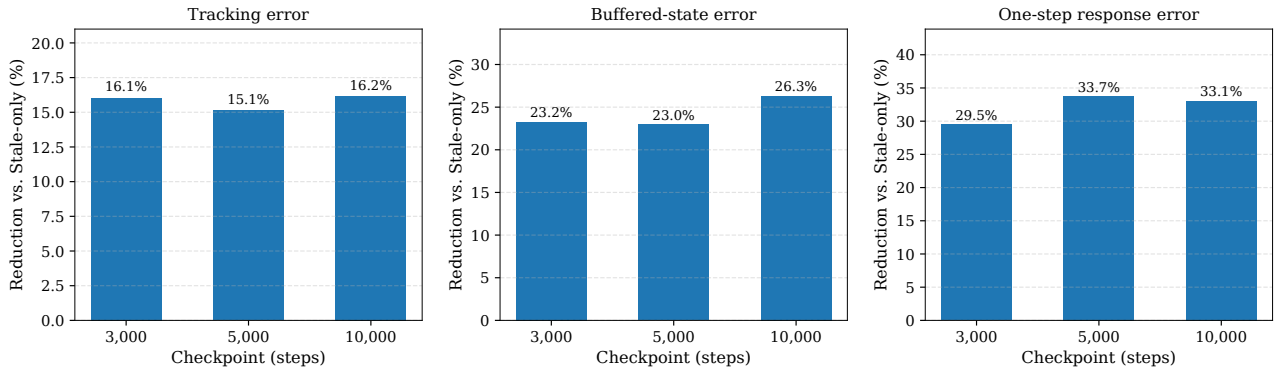


Fig. S2. Relative differences in tracking, buffered-state, and one-step response errors at the 3,000-, 5,000-, and 10,000-step checkpoints, calculated from the checkpoint mean values in Table S2.

nates. When $\mathbf{J}_M^{\text{eff}}$ is positive definite, $\lambda_{\min}$ characterizes its weakest spectral direction and $\kappa = \lambda_{\max}/\lambda_{\min}$ characterizes its condition number. The actual planner continues to use the original determinant rather than substituting a log-determinant optimization.

B. Geometric Snapshots and Spectral Distributions

The geometric analysis evaluates $M \in \{2, 4, 6, 8, 10\}$ AUVs in a 200×200 m workspace, with 1,000 independent geometric configurations per team size. AUV positions are sampled from the 10–189 m coordinate range subject to a minimum pairwise separation of 20 m. In this controlled diagnostic, the USV is placed at the AUV-team centroid and the vertical model reference is 100 m. The 5,000 evaluated configurations produce a maximum relative difference below 10^{-13} between the original determinant and Eq. (S5). The medians are already reproduced in Table V of the main manuscript; the figures here preserve the additional dispersion and distributional information that does not fit there.

The median minimum eigenvalue increases from 1.40×10^{-4} to 6.19×10^{-4} between 2 and 10 AUVs, while the median condition number decreases from 2.51 to 1.50. As shown in Figs. S3 and S4, favorable median trends are accompanied by a 5th-percentile minimum eigenvalue increase from 1.71×10^{-5} to 4.03×10^{-4} and a 95th-percentile condition-number reduction from 12.47 to 2.10. All 5,000 reconstructed matrices are positive definite in the reported sample. These are ensemble-specific observations and not a guarantee under arbitrary degenerate geometries.

C. Direct Coordinate-Perturbation Diagnostic

To test the determinant itself against small coordinate disturbances, the response applies three independent perturbation realizations to each geometric snapshot. Each AUV coordinate receives an independent uniform offset in $[-0.2, 0.2]$ m while the candidate USV position remains fixed. This gives 15,000 perturbed evaluations of the unchanged scalar objective. Table S3 records the computed absolute relative determinant changes.

Both the median and 95th-percentile relative perturbation change decrease across the sampled team sizes. Together with the spectral plots, this diagnostic indicates that the original determinant does not become progressively more sensitive to the imposed small coordinate disturbances within these sampled ensembles. The geometric snapshots do not replace an assessment of optimizer convergence, processor-level latency, or trained closed-loop operation with ten AUVs; the latter claims are not made here. Related informative path-planning work emphasizes that an information objective and the cost of obtaining informative positions must be assessed together [5]. Their computational implications are discussed in Section X.

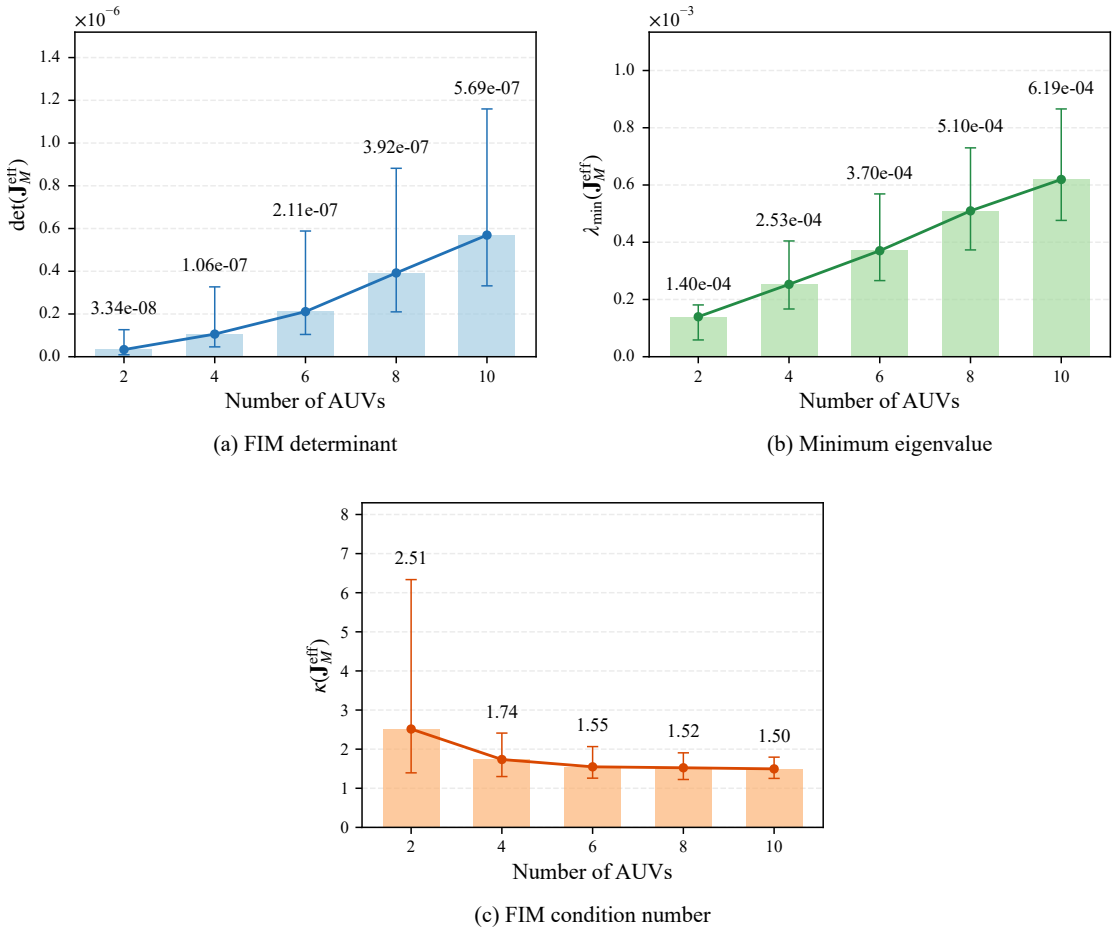
TABLE S3
ABSOLUTE RELATIVE CHANGES OF THE ORIGINAL DETERMINANT UNDER ±0.2-M AUV-COORDINATE PERTURBATIONS.

AUVs	Median change	95th percentile
2	0.181%	0.832%
4	0.141%	0.486%
6	0.115%	0.368%
8	0.099%	0.308%
10	0.087%	0.267%

IV. Technical Novelty and Formulation-Level Comparison

A. Distinction from the Earlier FIM–RL Framework

The technical contributions of the proposed method are best understood in relation to the authors' earlier FIM–RL USV–AUV framework [6]. That study already combined FIM-driven USV positioning with TD3-controlled underwater data collection and examined operation under challenging sea conditions. Neither FIM geometry optimization nor the use of TD3 in multi-AUV missions is, by itself, claimed as new here. The current problem is more specific: the surface leader operates on a slower update clock and must choose sensing-support positions using follower information that may be delayed, stale, or missing. Unlike geometry evaluation based on the currently available follower configuration alone, the stage-wise leader criterion in Eq. (25) of the main manuscript evaluates the FIM at the one-step follower-response configuration defined in Eqs. (41)–(43), with an additional squared USV-displacement penalty.



(a) FIM determinant

(b) Minimum eigenvalue

(c) FIM condition number

Fig. S3. Geometric scaling under the original FIM determinant and its equivalent 2×2 matrix: (a) determinant, (b) minimum eigenvalue, and (c) condition number. Error bars indicate 16th–84th percentile intervals over 1,000 geometric configurations per team size.

Specifically, at each scheduled leader update, the latest USV-accessible follower configuration $\bar{\mathbf{P}}_{a,t}$ is used to construct the predicted policy response $\mathbf{P}^{\mathrm{br}}_{a,t+1}(\mathbf{p}_u; \bar{\mathbf{P}}_{a,t})$ for a candidate USV position $\mathbf{p}_u$. The response map is formally candidate-conditioned through its state argument, but it need not yield distinct follower actions for two candidates that produce identical policy-state inputs. In all cases, the FIM evaluation geometry and leader motion penalty remain dependent on the candidate USV location. Periodic leader updates, leader-mediated buffered information, and the follower communication phase $\phi_t = (t \bmod N_u)/N_u$ together specify the information and timing structure absent from the earlier FIM–RL formulation. The communication model describes a simplified control-plane observation process, not an independently validated acoustic modem or a new physical-layer model.

B. Comparison with Related Coordination Paradigms

To distinguish these contributions from established approaches, Table S4 reproduces the formulation-level comparison developed for Response 3.1. The comparison includes representative hierarchical multi-AUV reinforcement learning [7], event-triggered leader–follower AUV communication [8], and AoI-aware multi-AUV MAPPO [9]. Unequal-rate hierarchical control is itself a well-established design choice [10]; the distinction in the present study is the coupling of this update schedule to buffered acoustic state access, one-step policy-response geometry evaluation, and phase-conditioned follower execution. Entries labeled as method families summarize their representative decision problems, rather than exhaustively characterizing every possible realization of those families.

The comparison has several important qualifications. Asynchronous MARL concerns decisions or learning with unsynchronized updates, but is not inherently a USV sensing-location planner. Event-triggered methods, including the AUV formulation in [8], adjust communication or control in response to a state-dependent event; here the leader update schedule is periodic rather than event-triggered. AoI-aware data collection can explicitly represent the age of received information [9], whereas the proposed phase variable measures the position within the leader's deterministic update cycle and is not the age of any particular packet. Communication-aware MPC can optimize multiple future controls over a receding horizon; the present leader instead optimizes a two-dimensional, stage-wise FIM–motion surrogate, informed by one-step follower-policy predictions. Therefore, the proposed framework does not claim a new general-purpose asynchronous MARL algorithm, AoI scheduler, event trigger, or MPC optimizer.

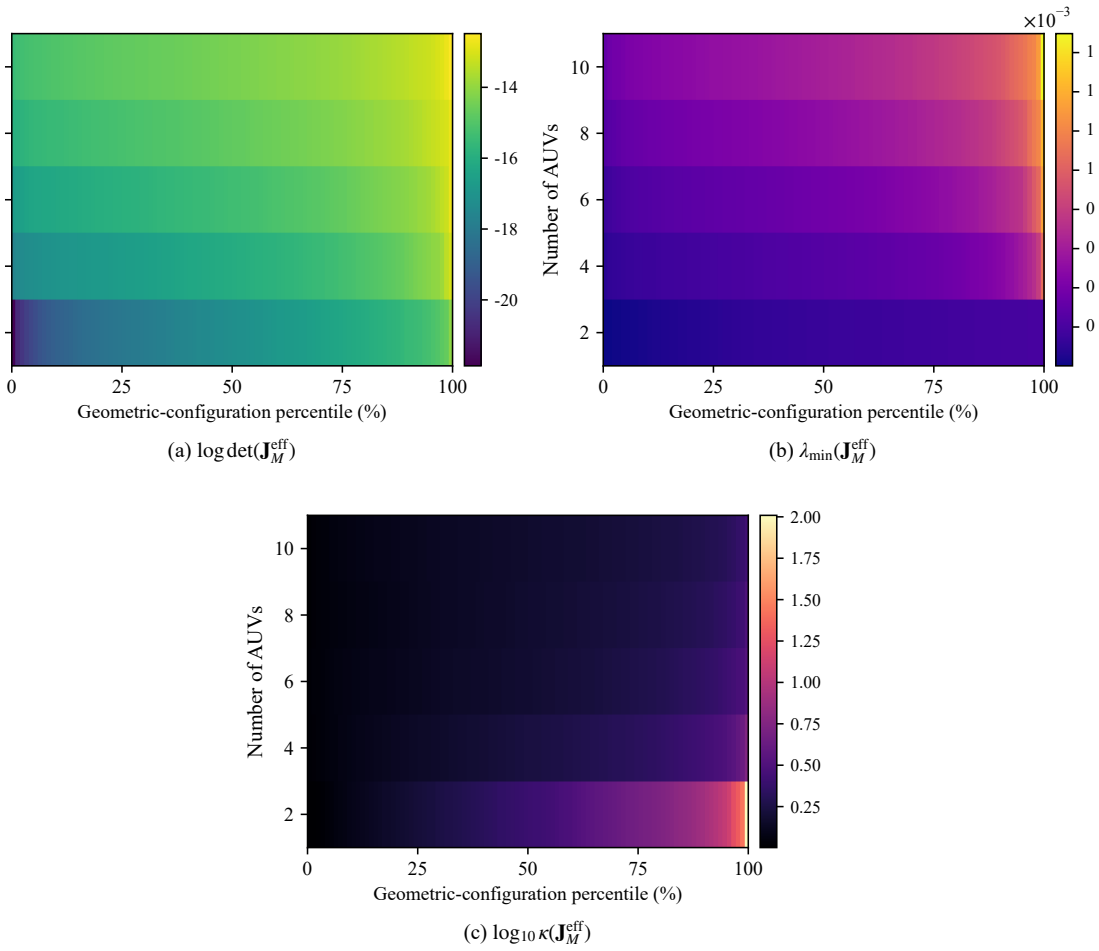


Fig. S4. Empirical percentile distributions of (a) log determinant, (b) minimum eigenvalue, and (c) log condition number. Each metric is independently ordered by configuration percentile, rather than plotted over simulation time.

TABLE S4
FORMULATION-LEVEL COMPARISON FOR CLARIFYING THE TECHNICAL NOVELTY IN COMMENT 3.1.

Work / Paradigm	**UF**	**SB**	**PR**	**AS**	**PH**	**GC**	**MR**	**AoI**	**ET**	**RH**	**Primary decision problem and technical distinction**
Earlier FIM–RL framework [6]	✓	△	×	×	×	△	×	×	×	×	FIM-based USV positioning with TD3 AUV task execution; does not formulate buffered-state one-step follower forecasting for use in the candidate-USV leader objective.
Asynchronous MARL (method family)	×	△	×	✓	△	△	×	△	△	×	Coordinates agents with nonsynchronous observations or updates; its defining decision is not the location of an information-providing FIM-based USV leader.
Hierarchical MARL [7]	×	×	×	△	×	×	×	×	×	×	Uses hierarchical policies for multi-AUV tracking; the hierarchy is not a stale-information-conditioned mobile sensing-geometry planner.
Event-triggered AUV formation [8]	×	△	×	✓	×	△	×	△	✓	×	Determines when to transmit or update via event conditions and dual prediction; does not select FIM-optimal USV positions after policy-predicted follower motion.
AoI-aware multi-AUV MAPPO [9]	×	△	×	△	×	△	×	✓	×	×	Optimizes data freshness and energy during AUV collection; packet AoI differs from the phase of a periodic USV update cycle.
Comm.-aware MPC (method family)	×	△	×	△	×	△	△	△	△	✓	Optimizes a sequence of future control actions under communication constraints, rather than a one-step follower-response FIM surrogate with learned AUV followers.
Proposed framework	✓	✓	✓	✓	✓	✓	✓	×	×	×	**Plans the USV sensing location from buffered leader-accessible states using policy-predicted AUV responses, geometry-linked delay/loss, periodic two-rate decisions, phase-conditioned RL, and motion regularization.**

Symbols: ✓ = explicit in the specified formulation; △ = related or partially represented; × = not explicitly included.
Dimensions: UF = FIM-guided USV with RL-based AUV execution; SB = stale/buffered leader information used for decisions; PR = one-step policy-based follower-response prediction used in candidate-USV FIM evaluation (identical policy inputs may yield identical predicted responses); AS = asynchronous or unequal-rate decision updates; PH = leader-update phase in the follower policy state; GC = geometry-coupled acoustic delay/loss affecting coordination information; MR = explicit leader-motion regularization; AoI = explicit packet-age state or objective; ET = event-triggered communication or control updates; RH = multi-step receding-horizon optimization.
Interpretation: × under PR does not imply that a method lacks all forms of prediction; PR refers specifically to prediction inside candidate-USV FIM evaluation. Method-family entries describe representative emphases, not every possible variant. The proposed framework models delayed/stale information but does not optimize packet AoI, event triggers, or an MPC horizon. The numbered citations in this comparison refer to the references at the end of this supplement.

C. Mechanistic and Empirical Interpretation

The formulation couples three links: USV–AUV geometry affects modeled delay and packet reception; available buffered estimates inform scheduled leader decisions evaluated using predicted follower positions; and the follower policies act at every step using USV-mediated state estimates and the current scheduler phase. The theoretical results in Section V of the main manuscript characterize selected properties of this stage-wise surrogate, including conditional buffered-error bounds, optimizer existence under stated regularity assumptions, prediction-error sensitivity, and a structural USV-motion limit. They do not prove general closed-loop optimality or unconditional robustness of learned policies.

The controlled experiments reported later in this supplement complement the formulation-level distinction. Table S7 shows that excluding phase augmentation increases the two-AUV TD3 tracking-error mean from 0.239 to 0.264 m, and excluding one-step prediction increases its diagnostic response mismatch from 0.92 to 1.14 m. The mobility-matched tests in Table S9 further compare the strategies under equal available travel caps, a common update interval, and zero motion regularization; at the 100-m cap and $N_u = 5$, mean tracking error is 0.2840 m for FIM and 0.2700 m for Proposed. These findings support an integrated, communication-aware coordination interpretation within the tested protocols. They do not establish additive component effects, exactly equal realized travel in the budget experiments, or universal superiority over every related paradigm.

V. Empirical Alignment of the Leader Surrogate and Task Return

The main manuscript distinguishes the system-level objective in Eq. (20) from the computationally tractable leader surrogate in Eq. (25). The former combines throughput, overflow, safety, AUV energy, tracking, and USV travel; the latter combines an FIM-based sensing-geometry proxy with squared leader displacement. The two objectives are related through the leader's effect on information quality and the followers' subsequent decisions, but they need not coincide numerically. We therefore evaluate agreement in candidate-level preferences, rather than requiring equality of the two objective values.

At matched leader-decision snapshots, each candidate USV position $\mathbf{p}_u$ is evaluated using the one-step follower-response configuration $\mathbf{P}^{\mathrm{br}}_{a,t+1}(\mathbf{p}_u; \bar{\mathbf{P}}_{a,t})$ defined by the policy-state and transition maps in Eqs. (41)–(43) of the main manuscript. The leader score combines the FIM evaluated at this predicted configuration and candidate-dependent motion regularization; a weighted five-step cycle return provides the task-level reference. The response map is formally candidate-conditioned, although candidates with identical follower-policy inputs may yield identical predictions. Spearman's ρ measures ranking agreement. A second diagnostic compares the unregularized FIM determinant with the portion of the five-step return excluding direct USV-motion cost. These comparisons answer different questions and should be read together.

TABLE S5
Measured candidate-ranking Spearman coefficients.

AUVs	Leader score vs. cycle return	FIM vs. non-motion return
2	0.885	0.446
3	0.852	0.472
4	0.817	0.438

Table S5 and Fig. S5 show regularized-score correlations of 0.885, 0.852, and 0.817 for the 2-, 3-, and 4-AUV settings, respectively. The FIM–non-motion correlations are more moderate at 0.446, 0.472, and 0.438. Thus the complete online leader criterion is more strongly associated with the cycle-level ranking than the FIM term is with the non-motion component in the evaluated candidate sets. The difference is compatible with an important influence of the motion regularizer on ranking agreement.

A separate original-environment action audit covers 36 leader-decision snapshots. The regularized leader objective selects the hold-position candidate in 30 snapshots and a nonzero-repositioning candidate in the remaining six. Thus, the objective favors remaining in place in most audited states but also selects repositioning under some conditions. These decision-level observations do not characterize the USV's accumulated travel over complete 1,000-step episodes, which is evaluated separately. Together with the ranking correlations, the audit indicates a predominantly conservative but not uniformly stationary action preference. Neither result independently isolates the contribution of the FIM term without checking the numerical magnitudes of the geometry and motion-penalty terms, nor does it guarantee identical candidate rankings for every state.

VI. Component-Wise Simulation Ablations

A. Controlled Two-AUV Protocol and Interventions

The aggregate matched-reference evaluation in the main manuscript changes several architectural choices simultaneously. This motivates separating the effects of scheduling, response prediction, and network-mediated observations rather than attributing an integrated gain to any single module [2], [3]. We therefore report the five component-wise experiments that were developed to distinguish their respective effects: time-scale separation, one-step follower-response prediction, phase-aware policy conditioning, motion regularization, and communication-aware observations. The controlled setting uses one USV, two TD3-controlled AUVs, 30 sensor nodes, and a 1,000-step horizon. Full and each of the five ablated variants are evaluated over 20 independent simulation repeats per configuration using the same evaluation seeds, task settings, metric definitions, and aggregation protocol. These matched-protocol means support direct descriptive component-wise comparisons; statistical significance is not inferred without run-level uncertainty analysis. The nominal geometry, delay model, 20-kHz carrier, and 4,096-bit packet length follow the main study.

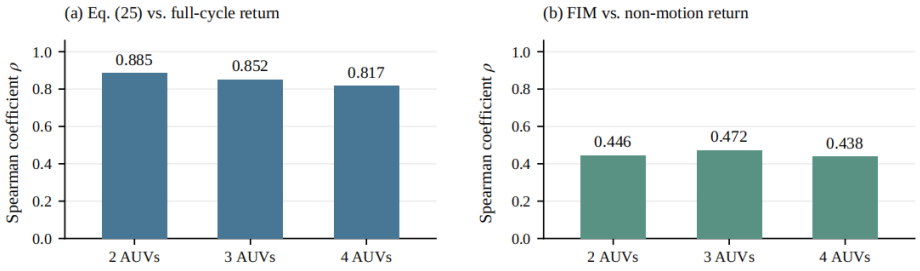


Fig. S5. Candidate-ranking agreement between the regularized leader objective and the weighted five-step system return (left), and between the FIM determinant and the non-motion return (right) for 2–4 AUVs.

The Full configuration uses the original leader determinant criterion, $N_u = 5$, and differential evolution with population parameter $P = 20$ and iteration limit $I = 100$. For leader-side interventions, the follower actor checkpoints are held fixed relative to Full. When changing the follower's input structure (phase or communication-aware observations), the relevant Full and modified policies instead undergo matched training and are evaluated with comparable seeds and budgets. Matched seeds do not imply identical realized acoustic receptions: robot trajectories still determine the geometry-dependent packet-success process. Table S6 specifies the interventions.

The tracking metric is evaluated using the post-communication position estimate available to the controller rather than a generated but unreceived measurement. Buffered-state mismatch and one-step response discrepancy are separate diagnostics. Sum rate refers to the *task-level data-collection* quantity in Mbps, not an acoustic modem bit rate. One scheduling caveat matters: with $N_u = 1$, the phase feature becomes constant automatically. The w/o time-scale variant is therefore a scheduler-level intervention, while the dedicated w/o phase comparison holds $N_u = 5$ fixed to examine phase conditioning more directly.

B. Cross-Metric Results and Mechanism-Specific Effects

Table S7 summarizes the mean outcomes of Full and the five ablations under the common 20-repeat statistical protocol. The Full tracking-error and USV-travel means (0.239 m and 0.910 km at the displayed precision) also agree with the corresponding two-AUV TD3 benchmark values in the main manuscript (0.2386 m and 0.91 km). The same evaluation and aggregation rules are used for all configurations. We report descriptive differences without asserting statistical significance in the absence of the underlying run-level uncertainty analysis.

Fig. S6 emphasizes that the complete framework does not minimize every outcome in isolation. Updating at every step reduces tracking error slightly (0.233 versus 0.239 m) and raises the reported sum rate (10.98 versus 10.87 Mbps), but increases USV movement from 0.910 to 3.420 km. Eliminating regularization similarly permits 1.090 km rather than 0.910 km of travel, while slightly reducing tracking error. Thus, time-scale separation and regularization mainly control the USV mobility trade-off, rather than necessarily producing the smallest tracking error.

In contrast, removing one-step prediction increases prediction mismatch from 0.92 to 1.14 m and tracking error from 0.239 to 0.253 m. Removing the phase feature at fixed $N_u = 5$ increases tracking error to 0.264 m, buffered-state mismatch to 0.99 m, and reduces the task sum rate to 10.58 Mbps. Removing communication-aware observation fusion increases tracking error to 0.267 m, buffered mismatch to 1.13 m, and one-step error to 1.27 m; the data-collection sum rate falls to 10.04 Mbps. These patterns distinguish the information-related mechanisms from the two movement-limiting mechanisms.

For a lower-is-better metric, the relative Full improvement over an ablation is $100(x_{\mathrm{abl}} - x_{\mathrm{Full}})/x_{\mathrm{abl}}$; for a higher-is-better metric, it is $100(x_{\mathrm{Full}} - x_{\mathrm{abl}})/x_{\mathrm{abl}}$. Applying these definitions to one mechanism-oriented metric per variant yields Table S8 and Fig. S7. Percentages in this display do not sum to a total contribution, since ablations are conditional interventions with interacting effects.

The mobility values can also be checked against the bounded leader-search region. With coordinatewise local half-width $r_u = 4$ m and $N_u = 5$, a 1,000-step episode allows at most approximately $200\sqrt{2}(4) \simeq 1.13$ km of successive USV displacement. The reported $N_u = 5$ variants fall below this bound; the $N_u = 1$ intervention naturally admits more movement opportunities. Overall, these component comparisons support complementary responsibilities for scheduling, predictive leader evaluation, phase augmentation, motion regularization, and communication-aware state handling without treating their effects as independently additive.

VII. Tracking under Matched USV Mobility Allowances

A. Common Movement Caps and Update Schedules

The principal benchmark reports a large reduction in USV travel for the integrated framework; in the two-AUV TD3 case the values are 3.59 km for the synchronous FIM reference and 0.91 km for Proposed. Because the proposed system moves only at scheduled leader updates and explicitly penalizes displacement, that travel reduction is partly structural. Unequal-rate control and information-seeking path planning provide additional reasons to examine tracking quality separately from available motion opportunities [5], [10]. A separate resource-controlled experiment therefore compares tracking while matching the update opportunities and allowable total movement.

TABLE S6
CONTROLLED COMPONENT ABLATIONS RELATIVE TO THE FULL FRAMEWORK.

Configuration	Intervention	Follower policy
w/o time-scale	Change $N_u = 5$ to $N_u = 1$, retaining the original FIM criterion, movement penalty, and acoustic observations. The scheduler-phase variable necessarily becomes constant.	Frozen
w/o prediction	Use buffered follower positions, rather than predicted one-step follower positions, in leader FIM evaluation.	Frozen
w/o phase	Retain $N_u = 5$ but mask the communication-phase feature from follower policy inputs.	Retrained
w/o motion regularization	Set $\lambda_u = 0$ while retaining the leader-update interval, FIM objective, and follower checkpoints.	Frozen
w/o communication observations	Remove delay-weighted follower-side fusion, retaining the packet-reception rule and held last-received information; never supply unreceived oracle positions.	Retrained

TABLE S7
MEAN CROSS-METRIC OUTCOMES FOR FULL AND FIVE CONTROLLED ABLATIONS, EACH EVALUATED OVER 20 INDEPENDENT SIMULATION REPEATS UNDER THE SAME STATISTICAL PROTOCOL.

Configuration	Tracking (m)	Buffered (m)	One-step (m)	USV travel (km)	Sum rate (Mbps)
Full	**0.239**	**0.82**	**0.92**	**0.910**	**10.87**
w/o time-scale	0.233	0.77	0.89	3.420	10.98
w/o prediction	0.253	0.91	1.14	0.955	10.74
w/o phase	0.264	0.99	1.04	0.942	10.58
w/o motion regularization	0.232	0.76	0.91	1.090	11.02
w/o communication observations	0.267	1.13	1.27	0.936	10.04

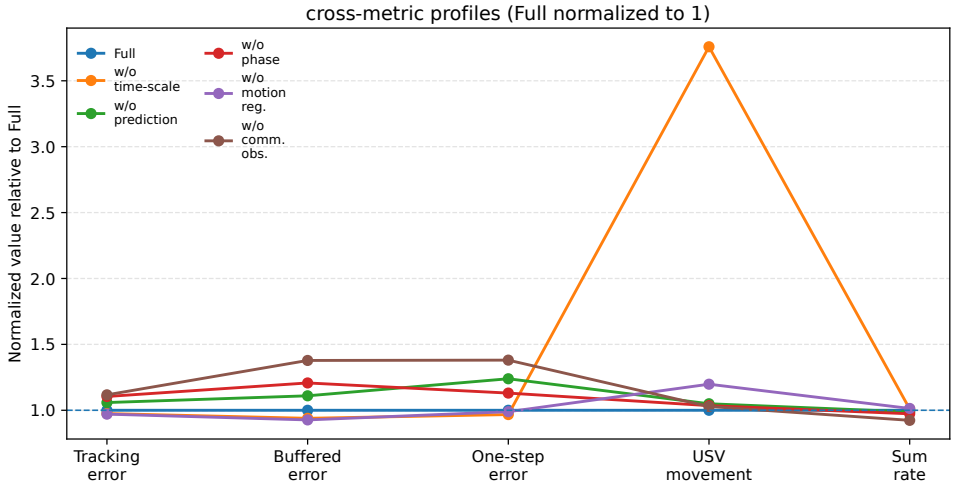


Fig. S6. Normalized cross-metric profiles for the complete method and five ablations. Each metric is normalized to its Full value of one; higher normalized tracking, buffered, prediction, and motion values are less favorable, whereas larger task-level sum rate is favorable.

TABLE S8
MECHANISM-ORIENTED DIFFERENCES COMPUTED DIRECTLY FROM TABLE S7.

Ablation	Target metric	Full	Ablation	Full improvement
w/o time-scale	USV travel (km) ↓	0.910	3.420	73.4%
w/o prediction	One-step error (m) ↓	0.92	1.14	19.3%
w/o phase	Tracking error (m) ↓	0.239	0.264	9.5%
w/o motion regularization	USV travel (km) ↓	0.910	1.090	16.5%
w/o communication observations	Sum rate (Mbps) ↑	10.87	10.04	8.3%

In the controlled two-AUV TD3 environment, both methods have identical initial USV positions, AUV and sensor-node initial conditions, coordinatewise local-search half-width $r_u = 4$ m, and the same cumulative travel cap $B \in$

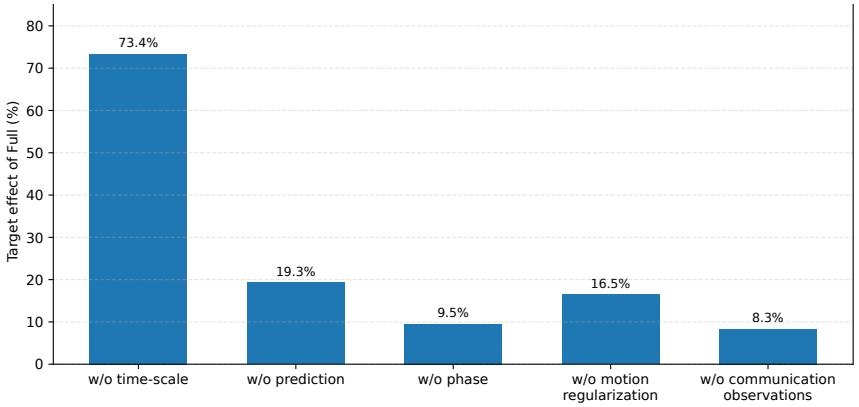


Fig. S7. Mechanism-specific relative changes with respect to the corresponding ablated variants. The selected outcomes are USV travel, one-step response error, tracking error, USV travel, and task-level sum rate, respectively.

$\{25, 50, 75, 100\}$ m. Both use $N_u = 5$ and $\lambda_u = 0$ in this auxiliary test so that neither method benefits from a stronger explicit movement penalty. All subsequent USV displacements, including the first relocation, count against the cap. The comparison therefore matches the *available* travel budget and update opportunities; it does not assert that the two realized path lengths are numerically identical on every run.

As shown in Table S9, both methods improve with a larger common travel allowance, and the proposed method has a lower reported mean tracking error for every tested cap. These mean differences alone do not establish statistical significance, particularly for the smallest caps. The whole-episode reductions increase from 0.5% at 25 m to 4.9% at 100 m. This comparison is distinct from the regularized 3.59-versus-0.91-km experiment in the main manuscript: matching N_u and setting $\lambda_u = 0$ deliberately change the auxiliary control conditions.

B. Tracking in the Terminal 250-Step Window

The resource-controlled comparison additionally reports a common terminal 250-step evaluation window under each mobility allowance. Its mean errors appear in the right-hand columns of Table S9. For the 50-m cap the error changes from 0.3280 to 0.3200 m, while at 100 m it changes from 0.3000 to 0.2800 m. The reductions across the four caps range from 0.6% to 6.7%.

The results are described as a terminal-window comparison. They can additionally be interpreted as a strictly stationary-tail comparison only for runs in which *both* USVs have zero displacement throughout the same 250 steps; an identical travel cap alone does not imply this condition. Even when both are stationary, their terminal positions and prior information histories need not match. Thus, these results characterize tracking under matched mobility allowances without asserting identical terminal sensing geometries.

C. Sensitivity to Matched Update Intervals

Finally, with both methods subject to the common 100-m travel cap, $r_u = 4$ m, and $\lambda_u = 0$, the update interval is varied jointly over $N_u \in \{1, 3, 5, 10\}$. Table S10 and Fig. S8 report the corresponding tracking errors.

Both methods obtain their lowest tracking error among the tested update intervals at $N_u = 5$, demonstrating a nonmonotonic dependence on the frequency of USV repositioning. Proposed exhibits lower error at all four frequencies, including $N_u = 1$ where the phase feature is identically zero. This describes lower mean errors for the evaluated integrated coordination strategy after controlling update opportunities and travel allowances. It does not establish equal realized travel, an independent causal gain from any single component, or significance of the smaller differences.

VIII. Communication-Model Robustness and Acoustic Sensitivity

This section reports controlled simulation experiments on the sensitivity of USV–AUV coordination to reception memory, shared impairment traces, forced outages, acoustic-model parameters, and limited message-service capacity. We distinguish leader-side buffered-state mismatch from task-level tracking error and interpret each comparison within its stated evaluation protocol.

A. Correlated Fading and Gilbert–Elliott Reception

The nominal control-plane model uses a geometry-dependent Bernoulli packet-success probability derived from path loss and Monte Carlo-averaged Rayleigh fading. As emphasized in the main manuscript, this approximation does not implement frequency-selective multipath, Doppler spread, explicit coding or retransmission, or collision-aware medium access. The experiments below therefore evaluate coordination robustness to specified packet impairments rather than physical modem or underwater acoustic-channel fidelity. GE models introduce temporally dependent reception states in networked estimation [11].

The stress experiments compare independent packet reception, correlated fading ($\rho = 0.90$), and correlated fading with GE reception ($P_{G\to B} = 0.08$, $P_{B\to G} = 0.12$). The configured aggregate packet-loss targets are 75%, 75%, and 82%, respectively. These percentages specify the stress-test

TABLE S9
TWO-AUV TD3 MEAN TRACKING ERROR (M) UNDER MATCHED $N_u = 5$, $r_u = 4$ M, ZERO MOVEMENT REGULARIZATION, AND IDENTICAL HARD USV TRAVEL CAPS.

Cap B (m)	Whole episode			Terminal 250-step window		
	FIM	Proposed	Reduction	FIM	Proposed	Reduction
25	0.3220	**0.3205**	0.5%	0.3400	**0.3380**	0.6%
50	0.3090	**0.3030**	1.9%	0.3280	**0.3200**	2.4%
75	0.2960	**0.2860**	3.4%	0.3140	**0.2990**	4.8%
100	0.2840	**0.2700**	4.9%	0.3000	**0.2800**	6.7%

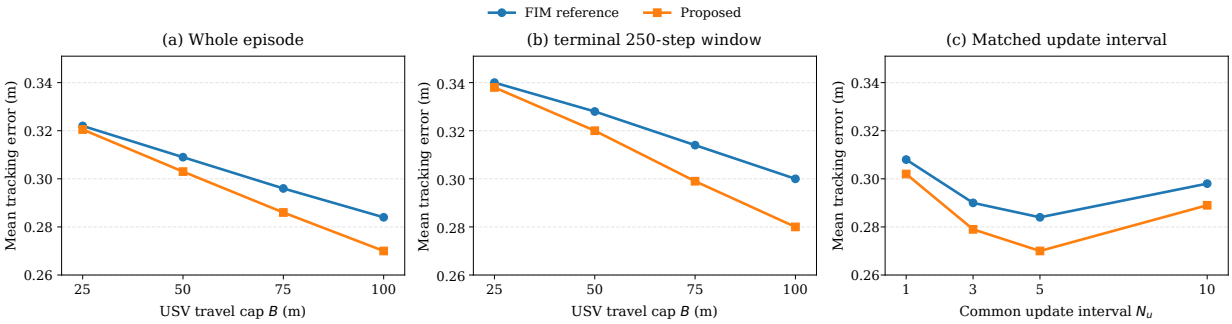


Fig. S8. Resource-controlled two-AUV TD3 tracking results: (a) whole-episode error under matched travel caps, (b) error in the common terminal 250-step window, and (c) performance under matched update intervals with a 100-m cap. A strictly stationary interpretation of panel (b) requires both USVs to remain motionless throughout the window, as verified by recorded movement rather than inferred from the cap.

TABLE S10
MATCHED-INTERVAL TRACKING ERRORS AT A COMMON 100-M USV TRAVEL CAP.

Common N_u	FIM (m)	Proposed (m)	Reduction
1	0.3080	**0.3020**	1.9%
3	0.2900	**0.2790**	3.8%
5	0.2840	**0.2700**	4.9%
10	0.2980	**0.2890**	3.0%

settings rather than physical modem measurements. The GE transition probabilities describe state persistence; the aggregate loss fraction also depends on the state-specific reception probabilities. Buffered-state mismatch is defined by Eq. (S1) and differs from both one-step response discrepancy and the task-level tracking error evaluated in the acoustic-parameter sweeps below.

For the three-AUV closed-loop comparison, Fig. S9 reports mean buffered-state mismatches of 2.42 versus 2.30 m for independent reception, 3.64 versus 3.34 m for correlated fading, and 5.35 versus 4.89 m for correlated fading with GE bursts (FIM versus Proposed). The corresponding reductions are 5.0%, 8.2%, and 8.6%. At the same aggregate loss target, the increase from independent to correlated reception is consistent with the greater impact of consecutive periods without refreshed information. The GE condition combines reception memory with a higher loss target, so its additional degradation should not be attributed to either factor alone.

The long-horizon evaluation reported earlier in this supplement and in Response 1.3 uses a different Proposed-versus-Stale-only protocol. The present FIM-versus-Proposed stress tests and that long-horizon comparison are separate evaluations with different baselines and sampling structures. Their buffered-state mismatch statistics are therefore interpreted within the respective protocols rather than treated as directly interchangeable results.

B. Matched Impairment Traces and Team Size

Trajectory-dependent strategies ordinarily experience different packet histories because their USV–AUV geometries differ. Paired-trace replay deliberately supplies a shared receive/drop indicator and delay history within each compared pair, reducing variation attributable to independent impairment draws. The trade-off is that this procedure suppresses part of the channel's native dependence on the different trajectories; it is a disturbance-isolation test rather than a full communication-model check.

Fig. S10 reports relative error reductions for $M \in \{2, 3, 4\}$ under the three reception processes. Seven of nine mean-error comparisons and eight of nine 95th-percentile comparisons favor Proposed. The three-AUV independent-reception and four-AUV correlated-fading cases favor FIM in mean buffered-state mismatch, while the four-AUV independent-reception case favors FIM at the 95th percentile. For correlated fading, the arithmetic average of the three separately calculated team-wise 95th percentiles decreases from 6.90 to 6.57 m (4.8% lower); under correlated fading with GE bursts, it decreases from 9.37 to 8.76 m (6.5% lower). These aggregates are *means of teamwise percentiles*, not percentiles computed from pooled samples.

The paired-trace results need not reproduce the performance ordering in Fig. S9. In particular, the three-AUV independent-

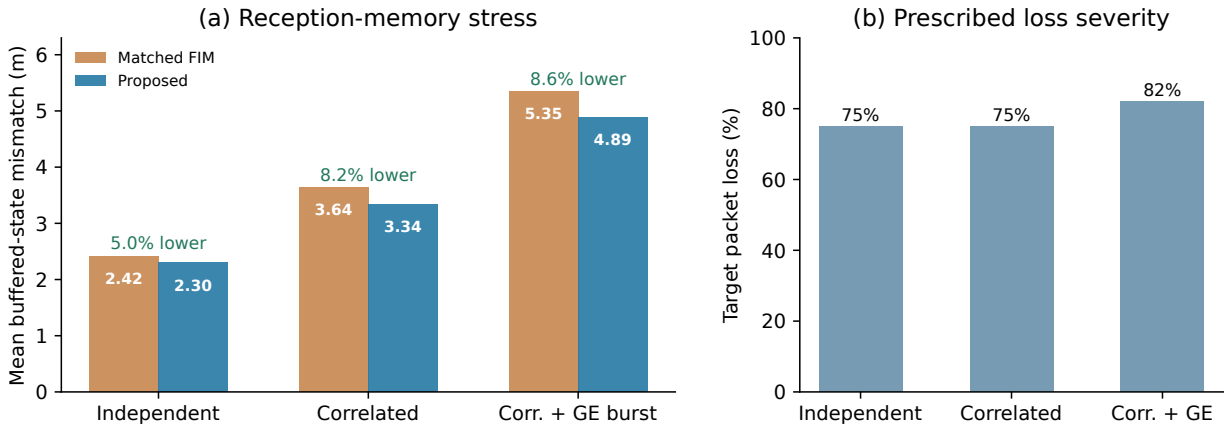


Fig. S9. Communication-memory sensitivity in the three-AUV simulation: (a) mean leader-side buffered-state mismatch under independent reception, correlated fading, and correlated fading with GE bursts; (b) configured packet-loss targets. Panel (a) reports buffered-state mismatch rather than task-level tracking error.

reception case favors Proposed in the geometry-dependent closed-loop evaluation but favors FIM in the matched-trace mean-error comparison. The two protocols differ in whether each strategy's trajectory determines its reception history. Their different rankings therefore characterize performance under different information-access conditions, rather than repeated estimates of a single common experiment.

C. Forced Communication Outages

To distinguish bursty reception from a controlled interval without successful updates, the blackout experiments suppress control-plane receptions for 5, 10, 20, or 40 consecutive *follower steps*. For $N_u = 5$, a 40-step blackout covers eight nominal leader-update intervals when aligned with the leader schedule. Thus, blackout duration is expressed in follower steps rather than the number of leader optimization calls. The leader retains its last received information throughout each blackout.

Fig. S11(a) reports time-averaged buffered-state mismatches of 2.45, 3.20, 4.26, and 6.58 m for FIM, compared with 2.37, 3.07, 4.10, and 6.75 m for Proposed at the four blackout lengths. Panel (b) reports a different statistic: the mean of runwise peak mismatches. These values are 3.65, 4.85, 7.20, and 11.55 m for FIM and 3.50, 4.62, 6.81, and 11.23 m for Proposed, corresponding to peak reductions of 4.1%, 4.7%, 5.4%, and 2.8%. At 40 steps, Proposed has a slightly *higher time-averaged mismatch* (6.75 versus 6.58 m; 2.6% higher) despite its lower mean peak. A lower mean of runwise maxima does not require a lower time average, so the conclusions remain metric-specific.

During a no-reception interval, buffered-state mismatch depends on the follower displacement that is not reflected in the retained information. Under the hold-last-value update, the loss-burst inequality $e^{\mathrm{buf}}_{k,t+d} \le e^{\mathrm{buf}}_{k,t} + \sum_{j=1}^{d} \|\mathbf{p}_{k,t+j} - \mathbf{p}_{k,t+j-1}\|_2$ relates the possible error increase to accumulated displacement rather than guaranteeing a small absolute error. The larger mismatches observed with longer blackouts are consistent with this dependence. The results characterize sensitivity over the tested outage durations; they do not imply recovery of undelivered packets or a uniformly small mismatch under arbitrary outages.

D. Source Level and Frequency–Packet-Length Sweeps

The acoustic-parameter experiments evaluate *task-level* AUV tracking error, rather than buffered-state mismatch, for three-AUV TD3 coordination over 1,000 follower steps. The leader update interval remains $N_u = 5$. Under correlated fading and SL $\in \{132.5, 135, 137.5, 140, 142.5\}$ dB, the mean tracking-error vectors, listed in increasing source-level order, are

$$\begin{aligned} \mathbf{e}_{\mathrm{FIM}} &= (0.345, 0.304, 0.279, 0.268, 0.262), \\ \mathbf{e}_{\mathrm{Prop}} &= (0.352, 0.285, 0.258, 0.246, 0.240). \end{aligned} \tag{S6}$$

The corresponding five-level averages are 0.2916 versus 0.2762 m for FIM and Proposed, respectively (5.3% lower for Proposed). Both methods exhibit decreasing tracking error as source level increases across the tested settings. However, the lowest source level favors FIM (0.345 versus 0.352 m), so the overall reduction does not imply an advantage at every source level.

The frequency–packet-length experiment covers the complete grid $f \in \{10, 15, 20, 30\}$ kHz and $L_p \in \{1024, 2048, 4096, 8192\}$ bits. At each tested carrier frequency, mean tracking error increases with packet length. This is an observed closed-loop trend within the evaluated configurations, rather than a general monotonicity guarantee of the packet-reception model. Thirteen of sixteen configurations favor Proposed, while the 30-kHz configurations with 2048, 4096, and 8192 bits favor FIM. The aggregate means are 0.3078 versus 0.2975 m (3.3% lower for Proposed). Averaging the sixteen configuration-specific percentage reductions gives 3.5% instead, because the individual percentages use different FIM denominators.

The highest-error configuration yields 0.3610 m for FIM and 0.3726 m for Proposed (3.2% higher for Proposed). The population standard deviation across the sixteen configuration means is 0.02165 m for FIM and 0.02980 m for Proposed (37.6% higher). This standard deviation describes cross-configuration variability, not uncertainty across repeated runs within a configuration. The lower across-setting mean for

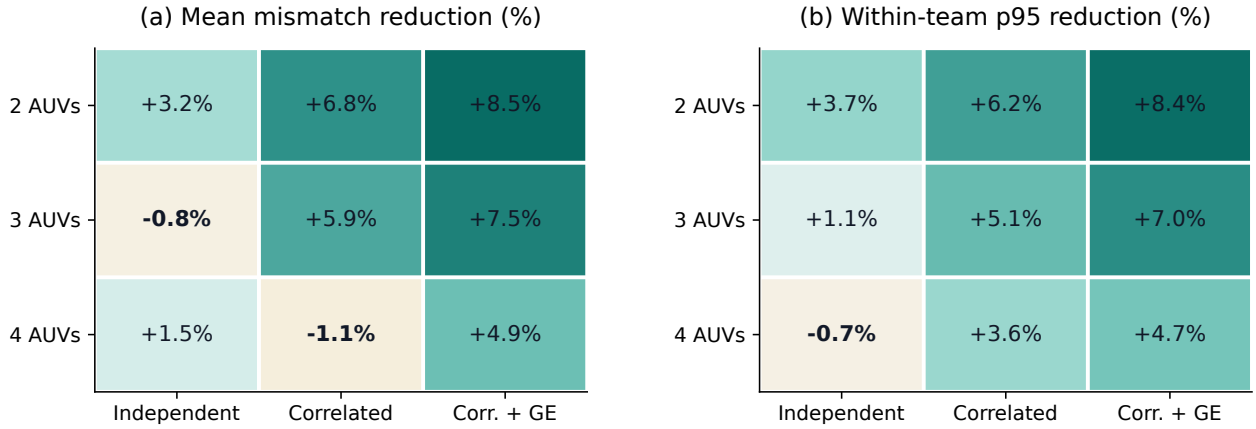


Fig. S10. Matched-impairment trace replay across 2–4 AUVs: relative reductions in (a) mean buffered-state mismatch and (b) within-team 95th-percentile mismatch. Each method pair receives the same reception indicators and delay sequence. Positive entries favor Proposed; negative entries favor FIM.

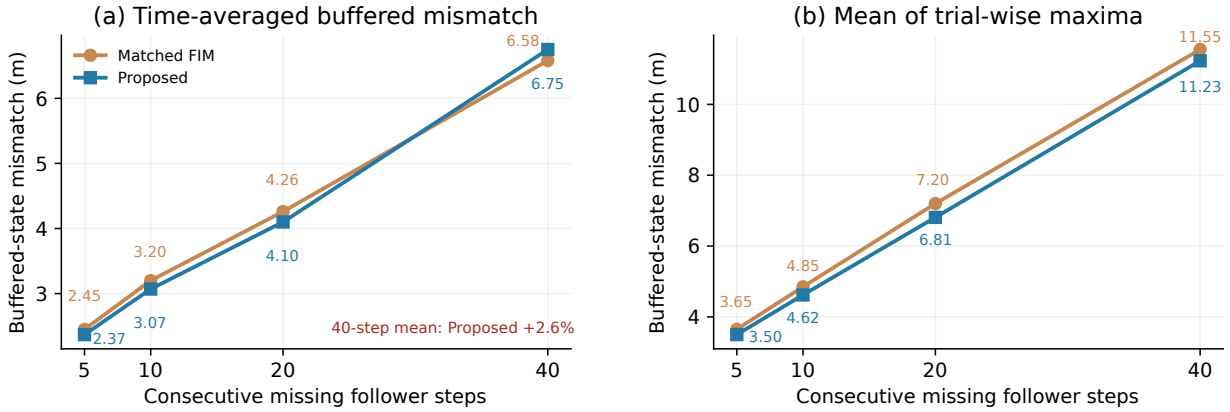


Fig. S11. Forced-outage simulation results: (a) time-averaged buffered-state mismatch and (b) mean of runwise peak mismatch for reception blackouts lasting 5–40 consecutive follower steps. The two panels summarize different aspects of error under interrupted communication.

Proposed therefore coexists with a higher worst-setting error and greater sensitivity across the tested acoustic configurations.

The source-level and frequency–packet-length sweeps share the acoustic-parameter setting $\mathrm{SL} = 135$ dB, $f = 20$ kHz, and $L_p = 4096$ bits. At this common setting, both report mean tracking errors of 0.304 m for FIM and 0.285 m for Proposed. These matching reported values should be read within the respective sweep protocols; common nominal acoustic parameters alone do not establish identical random seeds or packet histories.

E. Shared-Service Capacity and MAC-Layer Scope

A separate four-AUV simulation limits the number of control-plane messages served per follower step to one, two, or four. A physical underwater MAC additionally involves contention resolution, propagation-dependent collision timing, encoding, and retransmission [12], none of which is represented by a simple service cap. The FIM/Proposed mean tracking-error pairs are $0.371/0.382$ m at capacity one, $0.331/0.317$ m at capacity two, and $0.284/0.265$ m at capacity four. The most restrictive capacity favors FIM (3.0% higher error for Proposed), whereas capacities two and four favor Proposed (4.2% and 6.7% lower). These results identify a performance boundary under the tested shared-service constraint: the coordination advantage does not persist when the service budget is reduced to one packet per follower step. This is a coordination-level capacity result, not a collision-resolved MAC evaluation.

Together, the communication-stress experiments show lower mean or tail errors for Proposed in most of the tested comparisons, while identifying conditions under which that advantage is absent. In particular, the longest blackout yields a higher time-averaged buffered-state mismatch, and the lowest source level, selected 30-kHz configurations, and the one-packet service cap favor FIM in tracking error. These outcomes delimit the empirical coordination benefits under the adopted communication abstractions. The controlled packet-impairment results, task-level tracking results, and outdoor navigation measurements address complementary questions and are interpreted separately; none establishes physical acoustic-channel calibration or universal superiority under severe information constraints.

IX. Outdoor USV–AUV Prototype and Navigation Experiments

A. Field Site, Platforms, and Sensing Architecture

To complement the simulation findings with physical execution, outdoor trials were conducted in the waterbody shown

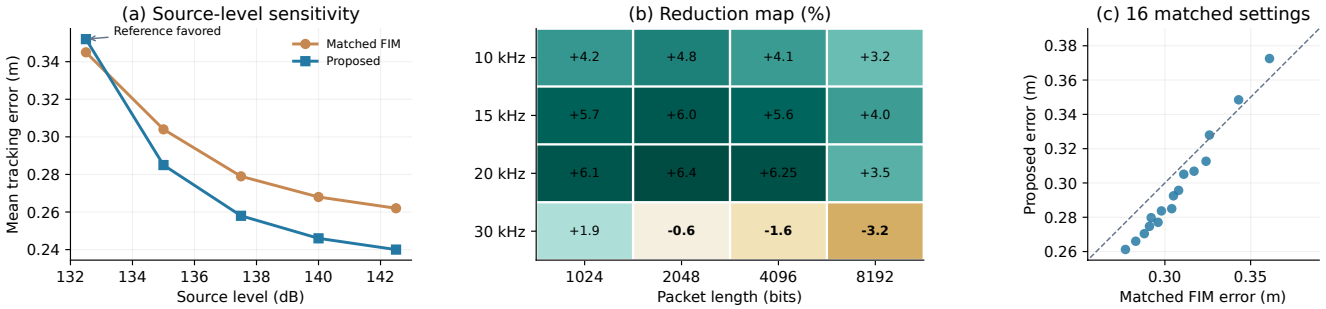


Fig. S12. Acoustic-parameter sensitivity of three-AUV TD3 coordination: (a) mean tracking error across source levels; (b) relative tracking-error reductions over the full frequency–packet-length grid; and (c) paired configuration-level tracking-error means. The three 30-kHz configurations favoring FIM are included.

TABLE S11
SUMMARY OF COMMUNICATION-SENSITIVITY SIMULATION RESULTS. BUFFERED-STATE MISMATCH AND TASK-LEVEL TRACKING ERROR ARE REPORTED SEPARATELY. PERCENTAGE DIFFERENCES ARE RELATIVE TO FIM AND INDICATE WHETHER THE PROPOSED ERROR IS LOWER OR HIGHER.

Experiment	Outcome / condition	FIM	Proposed	Difference
Channel memory	Mean buffered error, independent (m)	2.42	2.30	5.0% lower
	Mean buffered error, correlated (m)	3.64	3.34	8.2% lower
	Mean buffered error, correlated+GE (m)	5.35	4.89	8.6% lower
Trace replay	Mean teamwise p95, correlated (m)	6.90	6.57	4.8% lower
	Mean teamwise p95, correlated+GE (m)	9.37	8.76	6.5% lower
Forced outages	Mean of runwise peaks, 40 steps (m)	11.55	11.23	2.8% lower
	Time-averaged mean, 40 steps (m)	6.58	6.75	2.6% higher
Acoustic sweeps	Mean tracking over five source levels (m)	0.2916	0.2762	5.3% lower
	Mean tracking over 16 frequency–packet settings (m)	0.3078	0.2975	3.3% lower
Service capacity	Tracking, one packet/step (m)	0.371	0.382	3.0% higher
	Tracking, two packets/step (m)	0.331	0.317	4.2% lower
	Tracking, four packets/step (m)	0.284	0.265	6.7% lower

in Fig. S13. Representative site spans are approximately 75 and 120 m (Fig. S14), and the average water depth is approximately 5 m. The test fleet comprises one custom-built USV and three in-house-built, BlueROV-inspired underwater research robots operated as AUVs, as shown in Fig. S15. These are research-built platforms rather than off-the-shelf commercial BlueROV vehicles. The AUVs were commanded to operate at a depth of 3 m below the surface. The shallow field-test geometry and its real sensing and hydrodynamic conditions are distinct from the 100-m-depth-reference simulated environment. Thruster-driven AUV control has also been investigated using six-degree-of-freedom reinforcement learning [13]; the present field study instead evaluates goal-directed execution using the specified navigation-feedback architecture.

Each AUV was assigned a target horizontal coordinate and guided toward it using USBL-derived position feedback, with depth commanded to 3 m. Arrival was judged from the USBL-estimated position relative to the target. The same fleet and nominal navigation-task protocol were used for the FIM-guided reference and the proposed method, and *three repeated outdoor trials were performed for each method*. The numerical comparisons below are three-trial means, while trajectory maps and time series show representative runs. Fig. S17 records one 60-s execution at 0, 12, 24, 36, 48, and 60 s.

The instruments and measurements serve different purposes.



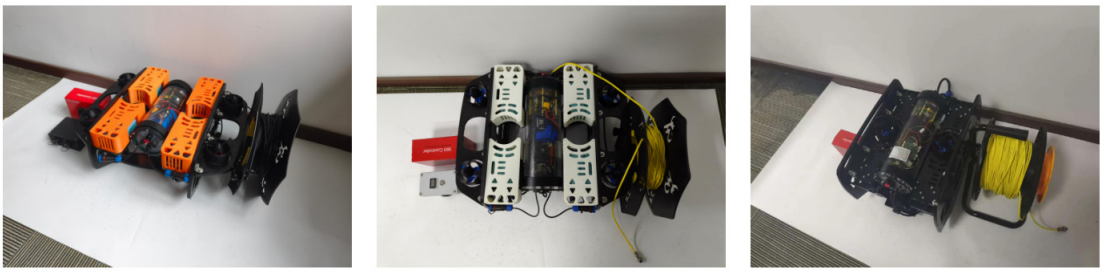
Fig. S13. Photograph of the outdoor waterbody used in the field evaluation.

USBL estimates supply position feedback for online goal-directed navigation; AUV traveled distance is reconstructed separately from IMU and DVL measurements fused through an extended Kalman filter (EKF). Consequently, the reported AUV route lengths are *not* obtained by summing successive USBL fixes. Asynchronous acoustic localization and estimation-assisted AUV tracking have been considered in related work [14], [15]; those references serve as context rather than direct validation of the particular IMU–DVL EKF



(a) Satellite view of the experimental site (b) Representative span of 75 m (c) Representative span of 120 m

Fig. S14. Satellite views of the field site with representative horizontal spans of approximately 75 and 120 m.



(a) Underwater platform 1 (b) Underwater platform 2 (c) Underwater platform 3

Fig. S15. The three in-house-built BlueROV-style underwater research platforms deployed as AUVs.



(a) USBL positioning equipment (b) DVL instrument

Fig. S16. Representative field instrumentation: (a) Ultra-Short Baseline (USBL) positioning equipment and (b) Doppler Velocity Log (DVL) instrument.

reconstruction used in these trials. This distinction reduces the direct influence of noisy acoustic fixes on the reconstructed path-length metric, while the actual steering behavior remains sensitive to the USBL feedback used by the controller.

B. Position Feedback and Motion-Response Interpretation

A first-order guidance relationship helps explain how acoustic position feedback can affect navigation. Denote the true horizontal AUV position by $\mathbf{p} = [x, y]^\mathsf{T}$, its USBL estimate by $\hat{\mathbf{p}} = \mathbf{p} + \boldsymbol{\varepsilon}$, and the goal by $\mathbf{p}_g = [x_g, y_g]^\mathsf{T}$. For a line-of-sight heading reference,

$$\psi_d = \operatorname{atan2}(y_g - \hat{y}, x_g - \hat{x}). \tag{S7}$$

For a small positioning discrepancy compared with the remaining goal distance $d = \|\mathbf{p}_g - \mathbf{p}\|_2$, the first-order heading perturbation is approximately

$$|\Delta\psi_d| \approx \frac{|\varepsilon_\perp|}{d}, \tag{S8}$$

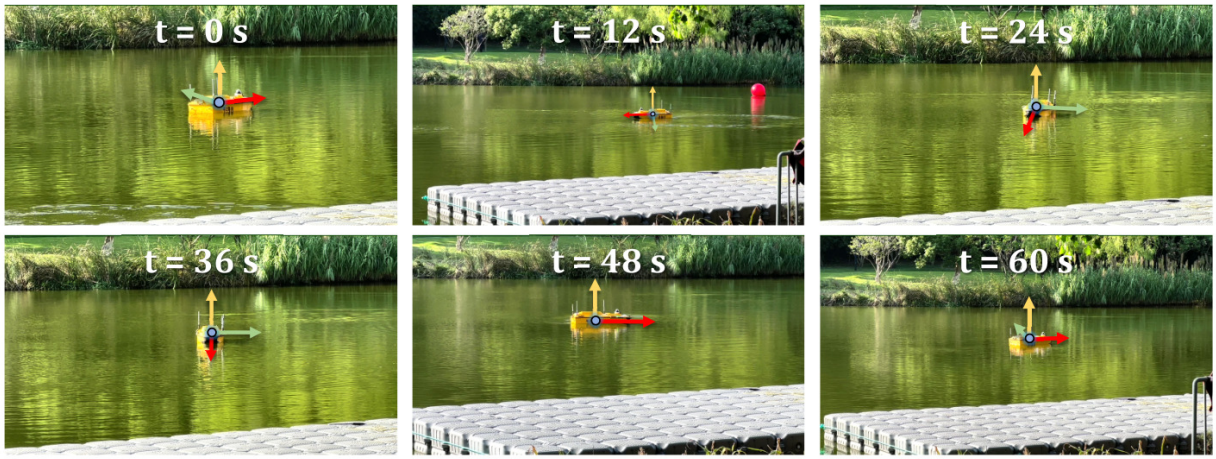


Fig. S17. Representative photographs at $t = 0$, 12, 24, 36, 48, and 60 s during a field navigation trial.

where $\varepsilon_{\perp}$ denotes positioning error perpendicular to the goal direction. The relation explains why noisy position feedback can induce heading corrections, particularly when the goal is close. It is a local guidance interpretation rather than a direct measurement of the USBL positioning-error distribution.

The proposed coordination architecture uses FIM-driven sensing geometry, buffered leader-side state information, renewed one-step follower-response estimates, and less frequent motion-regularized USV updates. Accordingly, field trajectories, reconstructed travel, depth, and yaw are meaningful *downstream navigation metrics* of this implementation. They do not separately identify the USBL positioning RMSE, since independently surveyed underwater ground truth was not available in the reported field protocol.

C. EKF-Reconstructed Trajectories and Traveled Distances

Let $\hat{\mathbf{p}}_{k,i}^{\mathrm{EKF}}$ denote the IMU–DVL EKF-estimated position of AUV k at sample i of one trial. Its estimated traveled distance is

$$L_{k,\mathrm{AUV}}^{\mathrm{EKF}} = \sum_{i=1}^{N_k-1} \left\| \hat{\mathbf{p}}_{k,i+1}^{\mathrm{EKF}} - \hat{\mathbf{p}}_{k,i}^{\mathrm{EKF}} \right\|_2 . \tag{S9}$$

For each method and AUV, the reported mean is computed across its three field trials. The USV travel distances are aggregated analogously from its recorded trajectory. These values measure accumulated movement rather than straight-line displacement between the starting and ending positions.

Fig. S18 shows goal-directed paths for the three AUVs in both conditions, with a visibly more compact USV path under the proposed framework. In the three-trial means, the proposed AUV routes measure 44.2, 96.9, and 77.1 m, compared with 45.4, 100.3, and 80.4 m for AUVs 1, 2, and 3 under the reference. These correspond to reductions of 2.64%, 3.39%, and 4.10%, respectively; the summed AUV mean travel decreases from 226.1 to 218.2 m (3.49%). More substantially, the recorded mean USV travel decreases from 91.0 to 38.5 m (57.69%). These field values are neither the simulation USV motion values nor an independent measurement of positioning accuracy.

TABLE S12
FIELD RESULTS FROM THREE OUTDOOR TRIALS PER METHOD.

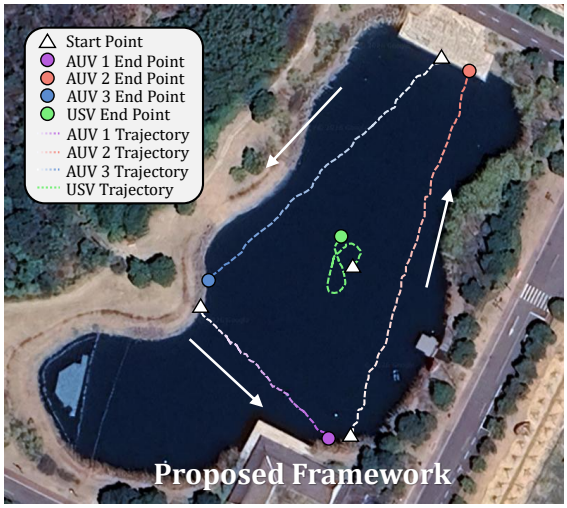
Quantity	FIM	Proposed	Reduction
AUV 1 path (m)	45.4	44.2	2.64%
AUV 2 path (m)	100.3	96.9	3.39%
AUV 3 path (m)	80.4	77.1	4.10%
Sum of AUV means (m)	226.1	218.2	3.49%
USV path (m)	91.0	38.5	57.69%
Depth SD (m)	0.07875	0.07530	4.38%
Yaw SD (degrees)	3.8063	3.4693	8.85%

D. Depth and Yaw Response

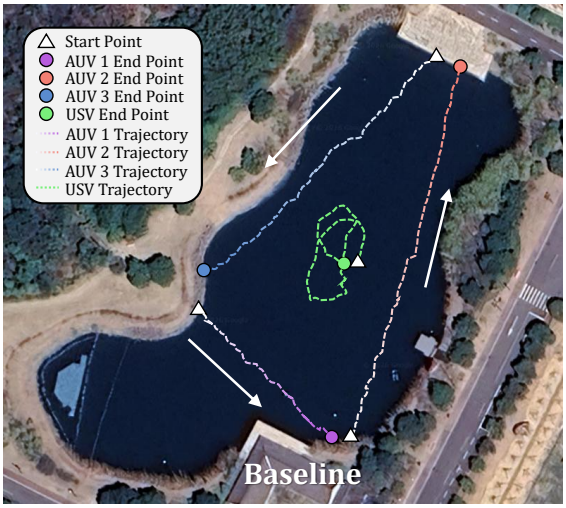
For the 60-s depth and yaw analyses, trial-wise standard deviations were first computed separately and then averaged over the three repetitions of each method. Figs. S19 and S20 provide representative time series and the resulting three-trial mean dispersions.

The mean depth standard deviation falls from 0.07875 to 0.07530 m, a 4.38% reduction. The corresponding mean yaw standard deviation decreases from 3.8063° to 3.4693°, an 8.85% reduction. The field measurements indicate maintained fixed-depth operation and a modest improvement in mean heading consistency, alongside the shorter AUV and USV trajectories. Table S12 summarizes the original three-trial mean values in one location.

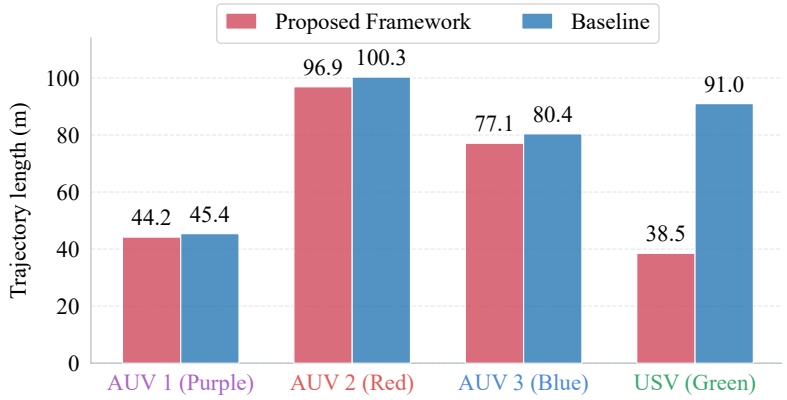
These field outcomes document execution of the proposed coordination architecture with physical vehicles and actual navigation feedback. The shorter AUV routes and lower yaw dispersion give indirect task-level support for more efficient USBL-feedback-driven navigation. However, environmental disturbances, vehicle dynamics, controller responses, and uncertainty of the IMU–DVL EKF reconstruction may contribute to the measured differences. Accordingly, these three-trial descriptive differences do not establish statistical significance, a ground-truth underwater position-error reduction, directly calibrated USBL RMSE, or empirical validation of the nominal Rayleigh/GE channel models used in the simulations.



(a) Proposed framework



(b) Baseline



(c) Trajectory-length comparison

Fig. S18. Representative field trajectories under (a) Proposed and (b) the FIM-guided reference, and (c) mean traveled distances over three repeated outdoor trials per method. AUV distances are reconstructed through IMU–DVL EKF odometry, not by summing USBL position fixes.

X. Computation, Acoustic Exchanges, and Deployment Resources

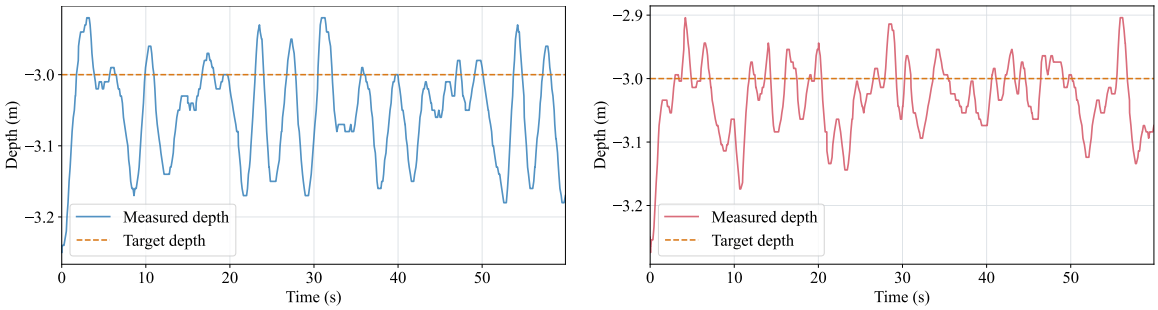
The additional experiments also motivate a separate assessment of resource requirements. A meaningful deployment analysis must distinguish analytical operation counts from processor-measured runtime and acoustic-message accounting from modem-measured throughput. The underlying robots and navigation instruments are the same for the proposed method and the reference in the field comparison, so their hardware base cost is largely shared.

A. Leader and Follower Online Computation

With M AUVs and per-follower policy-inference cost $C_\pi(M)$, the AUV inference work scales as $\mathcal{O}(MC_\pi(M))$ per follower step. For the original scalar FIM determinant, evaluating the M individual contributions and $M(M-1)/2$ pairwise contributions directly costs $\mathcal{O}(M^2)$ per candidate geometry. For a leader solver evaluating N_{eval} candidates at each scheduled update, a useful bound is therefore

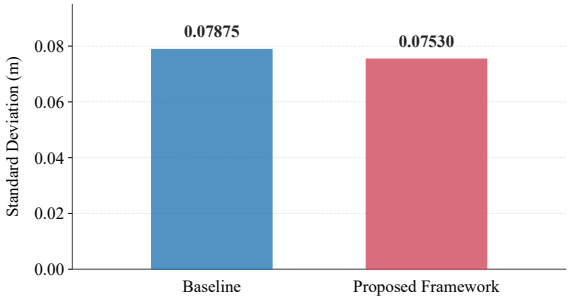
$$\mathcal{O}\left[N_{\text{eval}}\left(MC_\pi(M)+M^2\right)\right]. \tag{S10}$$

For the differential-evolution formulation specified in the main paper, the population size $P = 20$ and maximum iteration count $I = 100$ give a nominal population–iteration factor $PI = 2{,}000$ for workload analysis. A different solver, including any optional GPU-based multi-start optimizer, requires its own numerical settings and timing measurements and is not covered by this DE cost estimate. This factor is not asserted to equal a measured solver evaluation count, which depends on implementation details. Leader updates occur once every $N_u = 5$ steps, so the workload is periodically incurred rather than performed every follower step. This structural observation does not establish measured wall-clock computation latency on a particular onboard CPU. IMU–DVL EKF processing constitutes a separate navigation and estimation cost in the physical measurements.



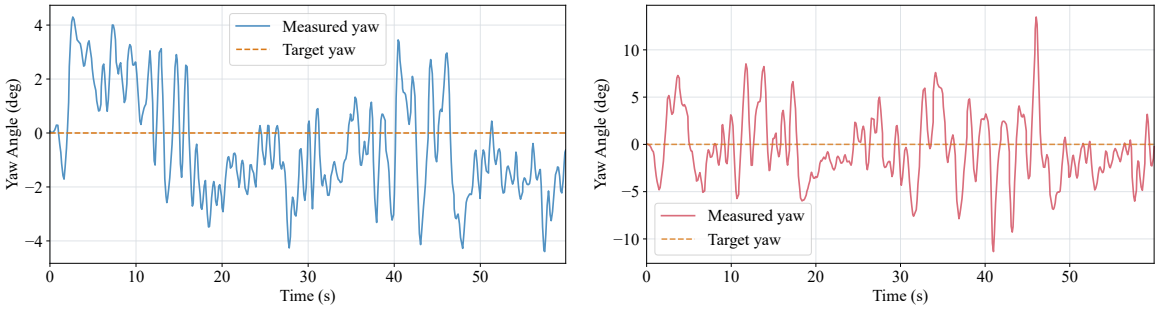
(a) Baseline

(b) Proposed framework



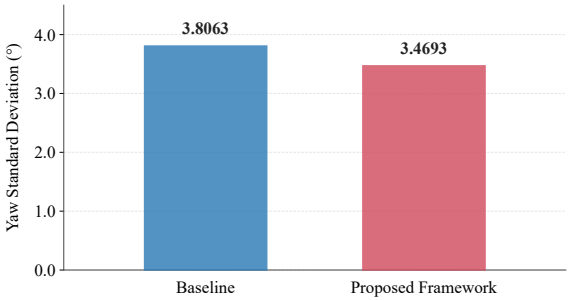
(c) Depth standard deviation

Fig. S19. Field depth responses at a commanded 3-m operating depth (approximately -3 m in the plotting sign convention): (a) FIM-guided reference, (b) Proposed, and (c) means of the three trial-wise depth standard deviations.



(a) Baseline

(b) Proposed framework



(c) Yaw standard deviation

Fig. S20. Field yaw responses: (a) FIM-guided reference, (b) Proposed, and (c) means of three trial-wise yaw standard deviations. Traces show representative executions.

B. Positioning Traffic and Propagation Time

Acoustic positioning exchanges must be distinguished from optionally packetized navigation-feedback or control mes-

sages. If AUV k requires positioning frequency f_k and q_k acoustic transmissions per update, the nominal fleet-wide positioning demand is

$$N_{\mathrm{tx}} = \sum_{k=1}^{M} f_k q_k \quad \text{transmissions/s}, \tag{S11}$$

prior to retries. Additional command types with frequencies g_j and payloads b_j bytes contribute $R_{\mathrm{ctrl}} = \sum_j g_j b_j$ bytes/s of nominal message payload. These expressions are accounting relationships, not modem data rates or measured communication logs. The leader's FIM evaluation of candidate geometries can reuse already received positions and need not require new USBL range exchanges for each numerical candidate.

For acoustic slant range r and propagation speed c, the one-way propagation-time component is $\tau_{\mathrm{prop}} = r/c$. Actual exchange latency additionally depends on processing, interrogation, and scheduling delays. The nominal simulator settings of 20 kHz and 4,096-bit packets are not specifications of the physical USBL equipment. Measured modem throughput, packet counts, and latency require separate synchronized acoustic records.

C. Hardware-Independent Cost Accounting

For one USV and M AUVs, a platform-independent cost decomposition is

$$\begin{aligned} C_{\mathrm{total}}(M) = {} & C_{\mathrm{USV}}^{\mathrm{base}} + M C_{\mathrm{AUV}}^{\mathrm{base}} + C_{\mathrm{USBL}} \\ & + C_{\mathrm{nav}}(M) + C_{\mathrm{compute}} \\ & + C_{\mathrm{integration}} + C_{\mathrm{operation}}. \end{aligned} \tag{S12}$$

The base-vehicle terms exclude instrumentation already itemized separately; the remaining terms cover USBL equipment, IMU/DVL and other navigation sensing, computing, calibration/integration, and field operation. Since the original fleet comprises custom research platforms, the response reports no universal commercial platform price. A reduction in traveled distance may benefit operations, but without measured propulsion power it should not be reinterpreted as a quantified energy or monetary saving.

XI. Summary of Supplementary Evidence

The formulation-level comparison distinguishes the proposed information and update structure from earlier FIM–RL and related coordination paradigms, without treating its component techniques as individually novel. The long-horizon checkpoints show sustained error differences over the sampled 10,000-step windows, and the geometric experiments diagnose the implemented FIM determinant as AUV team size increases without claiming a trained ten-AUV deployment. Candidate-ranking tests link the regularized leader score to a task-level cycle-return diagnostic, while controlled component and mobility-matched comparisons separate coordination benefits from the structural consequences of slower, movement-penalized leader updates. The communication-stress experiments evaluate correlated and bursty reception, matched impairment traces, forced outages, acoustic-parameter changes, and shared-service restrictions. Their results show advantages in most tested comparisons while documenting unfavorable conditions and metric-specific trade-offs. Finally, repeated outdoor tests demonstrate target-directed physical navigation with reduced mean movement and modestly lower depth/yaw variability, without treating these indirect metrics as measured USBL localization RMSE or physical acoustic-channel validation. These complementary experiments provide descriptive support within their respective protocols while preserving the distinctions among simulation metrics, numerical geometry diagnostics, and field-observed task performance. They do not by themselves establish candidate-specific actor recomputation, hardware-timing feasibility, universal statistical significance, or direct field calibration of underwater localization error.

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